

# Redirected Walks through Hazardous Environments: Detection Thresholds of Curvature Gains in 360 Video and 3D Telepresence Setups

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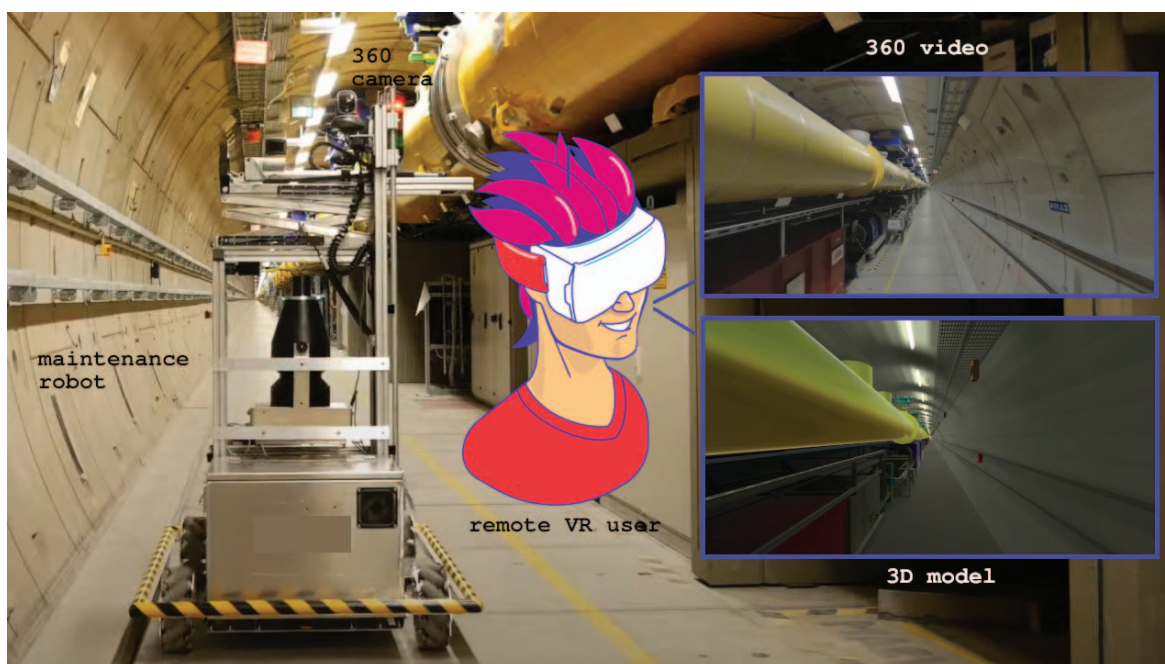


Figure 1: The setup shows a maintenance robot with a 360 camera in an accelerator tunnel and a remote VR user who can control the robot either by using the 360 video image or a 3D modeled environment as a visual stimulus. To navigate the robot, the VR user leverages redirected walking techniques.

## ABSTRACT

This paper investigates the application of redirected walking techniques in the context of a virtual reality (VR) telepresence system designed for remote control of a maintenance robot within hazardous environments. We estimate detection thresholds for curvature gains in two distinct environments: a 360-degree video-based VR setup and a three-dimensional (3D) modeled environment. Building upon prior work that focused on translation and rotation gains in 360 video-based VR, this research explores curvature gains in 360 video-based VR. Moreover, we compared results to detection thresholds of 3d modeled environments. Our findings reveal differences in detection thresholds between the 360-degree video and 3D modeled environments, with the latter displaying a larger shift of thresholds for curvature, and therefore offer insights into perceptual aspects and limitations of human sensory system.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

## 1 INTRODUCTION

In the landscape of virtual reality (VR) and telepresence technologies, the pursuit of creating immersive, intuitive, and efficient user experiences remains a constant challenge [18]. Telepresence systems, particularly those designed for remote control of robots in complex and hazardous environments, have seen significant advancements. One such environment is the accelerator tunnel of an electron laser research facility which was leveraged as application example in our study (see Figure 1).

As the demand for remote maintenance and operation of such facilities grows, so does the need for effective and natural methods of human control. Virtual reality, with its ability to provide a sense of presence and immersion [22], presents a promising solution. To navigate and control robots in such environments [21, 23], researchers have turned to techniques like "redirected walking" [19], a method where users perceive themselves to be moving straight while physically walking along curved paths. This approach has shown great potential in minimizing the physical space required for VR experiences while maintaining user comfort and the perception of vast virtual environments [15].

Previous research in this area has primarily focused on estimating detection thresholds for translation and rotation gains in 360-degree video-based VR environments [29]. These thresholds are critical for ensuring that users do not notice discrepancies between their

physical movements and their virtual counterparts, which can lead to discomfort or disorientation.

This research explores whether a 360-degree video can be used in conjunction with curvature gains or if a three-dimensional (3D) modeled environment offers superior results in terms of detection of redirection.

Both options have different advantages: A 360 video offers usually a more detailed view on the environment and does not require any 3D modelling work. So, when using a 360 camera on the robot that streams into a VR headset, changes in the real environment are directly visible. However, latency can be a problem depending on the connection availability. A 3D model, on the other side, is more undemanding regarding connection and bandwidth because it only requires 3d position data (e.g. from inertial measurements or GPS data) from the robot to display the correct part of the environment to the VR user. Moreover, it offers more flexible viewing positions and stereoscopic disparity. However, changes in the real environment need to be reflected manually in the virtual model which makes it more inflexible in this regard.

The motivation behind this study is twofold. First, we seek to determine the detection thresholds for curvature gains in a 360-degree video environment to close a gap in the literature. Second, we aim to compare these thresholds to estimations obtained within a comparable 3D modeled environment, which has not been done in the past as long as we know. This comparative analysis will shed light on the advantages and limitations of each approach and help guide future developments in telepresence technology.

Our findings indicate that the thresholds for curvature gains tend to differ between the 360-degree video and 3D modeled environments, with the latter demonstrating a higher shift of the detection thresholds (DTs) and point of subjective equality (PSE). These results have implications for the design and implementation of telepresence systems, particularly in scenarios like remote maintenance within accelerator tunnels, where precise control and user comfort are paramount. In the subsequent sections of this paper, we will delve into the methodology, experimental setup, and detailed results of our study.

## 2 RELATED WORK

Redirected Walking (RDW) is a collection of VR locomotion techniques that enables humans to walk on paths in the real world, which may vary from the paths they perceive in the virtual environment (VE) [3, 17, 27]. RDW can be implemented by manipulations applied to the virtual camera, usually by using *redirection gains*, which require the user to compensate by repositioning and/or reorienting themselves in order to maintain their intended walking direction in the VE. Within certain detection thresholds such manipulations are subtle for the user.

The traditional approach of RDW that was introduced by Razaque et al. [19] and extended by others [6, 7, 9, 25] is based on slightly rotating the user's view to one direction while she walks on a straight path in the VE. Then, she will unconsciously compensate the rotation by walking on a curved path in the opposite direction. As a result, she will walk on a circular arc in the real world (see Figure 2). When redirecting a user, her visual feedback is consistent with movements in the VE, but proprioceptive and vestibular feedback systems are coupled to the physical world. It is known from perceptual and cognitive psychology that vision often dominates proprioception and vestibular sensation when the information from these senses is in conflict [1, 5]. However, if the visual-proprioceptive conflicts are small enough, humans will not even be able to notice that they are manipulated. Since humans tend to compensate for slight visual-proprioceptive inconsistencies during locomotor control while walking, it becomes possible to guide users along real-world paths, which differ from the paths they perceive in the virtual world.

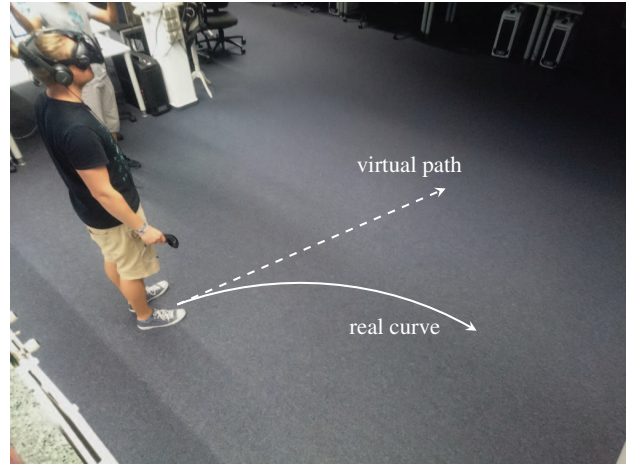


Figure 2: A VR user walking virtually straight while being redirected to the right using a curvature gain [14].

Steinicke et al. described such manipulations by means of gains. A gain describes differences between real and virtual motions, i.e., ratios between a user's movements in the real world and in the VE. For example, curvature gains can be defined as follows:  $g_C := \frac{1}{r_{real}}$ , where  $r_{real}$  describes the radius of the circle on which the users are guided in the real world [25]. A translation gain  $g_T := \frac{T_{virtual}}{T_{real}}$  manipulates the virtual velocity of the user, making her walk slower or faster in the VE compared to the velocity in the real world [11]. A rotation gain  $g_R := \frac{R_{virtual}}{R_{real}}$  increases or decreases the virtual rotation of the viewpoint relative to the rotation in the real world.

Steinicke et al. conducted several experiments to reveal detection thresholds for translation, rotation, and curvature gains [25]. The detection thresholds indicate just-noticeable differences between vision and vestibular system. According to their results, a straight path in the VE can be turned into a circular arc in the real world with a radius of at least 22m, for which users are not able to consciously detect manipulations. This means, that a manipulation of  $2.6^\circ/m$  is unnoticeable. These experiments have been replicated with different settings and extended several times [2, 4, 8, 13, 24, 26]. For example, Grechkin et al. found that a radius of only 12m can be sufficient for unlimited straight walking in a VE (manipulation of  $4.9^\circ/m$ ) [10] and Rietzler et al. found a threshold of  $5.2^\circ/m$  [20].

Zhang et al. estimated detection thresholds for translation and rotation gains in a 360 video-based telepresence system and found that their results were consistent with most previous findings in fully computer-generated VEs [29]. They used pre-recorded 360 videos instead of a real-time video stream from a telepresence system for convenience and to exclude factors like latency. They did not evaluate curvature gains and did not compare VEs directly in an experiment. In our experiment, we use a similar setup and focus on these parts that were missing in the original study.

## 3 EXPERIMENT

For estimation of detection thresholds of curvature gains in 360 video and 3D modeled environments, we conducted a psychophysical experiment based on a Two-Alternative Forced-Choice (2AFC) task. For this, we used the application scenario of a maintenance robot of an accelerator tunnel in a research facility.

### 3.1 Participants

The experiment was completed by 16 subjects (3 female and 13 male, ages 19–64 years,  $M = 31.94$ ). All of our participants had normal or corrected-to-normal vision.



Figure 3: The virtual environments shown to the participants: A 360 video (top) and a 3D modeled environment (bottom) of the same part of an accelerator tunnel of an electron laser research facility. During operations, the tunnel can only be maintained by robots because of radiation.

## 3.2 Materials

The experiment took place in a laboratory room at University. We instructed the participants to wear a Meta Quest 2 headset, which provides a resolution of  $1832 \times 1920$  pixels per eye with an approximately  $90^\circ$  horizontal field of view and a refresh rate of 90 Hz. The guardian was calibrated so that there was an available walking space of  $6\text{m} \times 5\text{m}$ . The participants received instructions on text slides presented on the head-mounted display. A Quest controller served as an input device via which the participants provided responses during the experiment.

For the experiment, we simulated the telepresence setup in the VR headset and did not incorporate a live connection to a maintenance robot. The VE was rendered using the Unity3D engine and showed either a 360 video or a 3D model of an (straight) accelerator tunnel of an electron laser research facility (see Figure 3). The 360 video was captured in the tunnel on a moving platform. It is implemented in Unity in a way that a movement of the user on the forward axis plays back the corresponding section of the video. Hence, for the user it feels like moving through a 360 video and she can control the speed. The 3d modeled environment shows the exact same part of the tunnel and is based on a CAD model that was optimized for use in realtime.

## 3.3 Methods

We used a  $7 \times 2$  full-factorial within-subjects experimental design (7 gains and 2 VE types). We tested 7 different gains  $g_C \in \{-9^\circ/\text{m}, -6^\circ/\text{m}, -3^\circ/\text{m}, 0^\circ/\text{m}, 3^\circ/\text{m}, 6^\circ/\text{m}, 9^\circ/\text{m}\}$ . A negative gain means that the participant is guided towards the left of the virtual path and a positive gain means that the participant is guided towards the right of the virtual path. A gain of  $0^\circ/\text{m}$  means that the participant is not manipulated and virtual and real path match one to one. Each gain was repeated 4 times. We tested 2 different VE types: a 360 video and a 3D modeled environment. In total, the participants completed  $7 \times 4 \times 2 = 56$  trials. All trials were randomized.

These concrete gains were chosen based on previous work as described in Rietzler et al. who reported detection thresholds around  $5.5^\circ/\text{m}$  for curvature gains and recommended to use gains that scale in a linear way with the perceived manipulation [20].

### 3.3.1 Procedure

Before the experiment, all participants filled out informed consent forms and received detailed instructions about the experiment and on how to perform the experimental task. Furthermore, they filled out a demographics questionnaire and the Simulator Sickness Questionnaire (SSQ) [12].

Then, participants mounted the headset and started the actual experiment. For each trial, participants were instructed to walk along the tunnel shown in the VE. When they reached the end of the path, the tunnel environment was faded out and they were in an empty environment that showed the 2AFC task question and they had to answer the displayed question using the controller (see Section 3.3.2).

Afterwards, participants were guided back to the start position in the laboratory by displaying a line on the ground. The next trial started once participants reached the start position and indicated that they were ready to start by pressing a button on the controller. Participants were allowed to take breaks between trials. After the VR session, participants filled out the SSQ again.

The total time per participant, including pre-questionnaires, instructions, experiment, and post-questionnaires, was about 60 minutes, out of which around 45 minutes were spent in VR.

### 3.3.2 One-Alternative Forced-Choice Task

To measure the amount of deviation that is unnoticeable, we used a standard psychophysical procedure based on the One-Alternative Forced-Choice (1AFC) task. This experimental method is a common procedure in RDW research, however it is often incorrectly called 2AFC task [25].

After walking a path in the VE while a curvature gain is applied, the following question is displayed to the participants: “*Did you physically walk left or right of the virtually displayed tunnel path?*”. The participants have to choose between one of two answer possibilities, usually if they walked “*left*” or “*right*” of the virtually straight path [25]. Answers like “I don’t know” are not allowed. Instead, the participants have to choose one option randomly and will be correct in 50% of the cases on average.

The curvature gain at which the participants detected the manipulated path correctly in 50% of the trials is taken as the PSE, at which the participants estimate the walked path as straight. As the gain decreases or increases from this value the ability of the participant to detect the difference between physical and virtual movement increases, resulting in a psychometric curve for the discrimination performance. When the participant’s answers converge to 100% respectively the 0% chance level, it is more likely that they can detect the redirection reliably. A threshold is the point of intensity at which participants can just detect a discrepancy between physical and virtual motion. However, stimuli at values close to thresholds will often be detectable. Therefore, thresholds are considered to be the gains at which the manipulation is detected only some proportion of the time. In psychophysical experiments, usually the point at which the curve reaches the middle between the chance level and 100% is taken as a threshold. Therefore, we define the DT for gains smaller than the PSE to be the gain at which the participant has 75% probability of choosing the “left” response correctly and the detection threshold for gains greater than the PSE to be the gain at which the subject chooses the “left” response in only 25% of the trials (since the correct response “right” was then chosen in 75% of the trials).

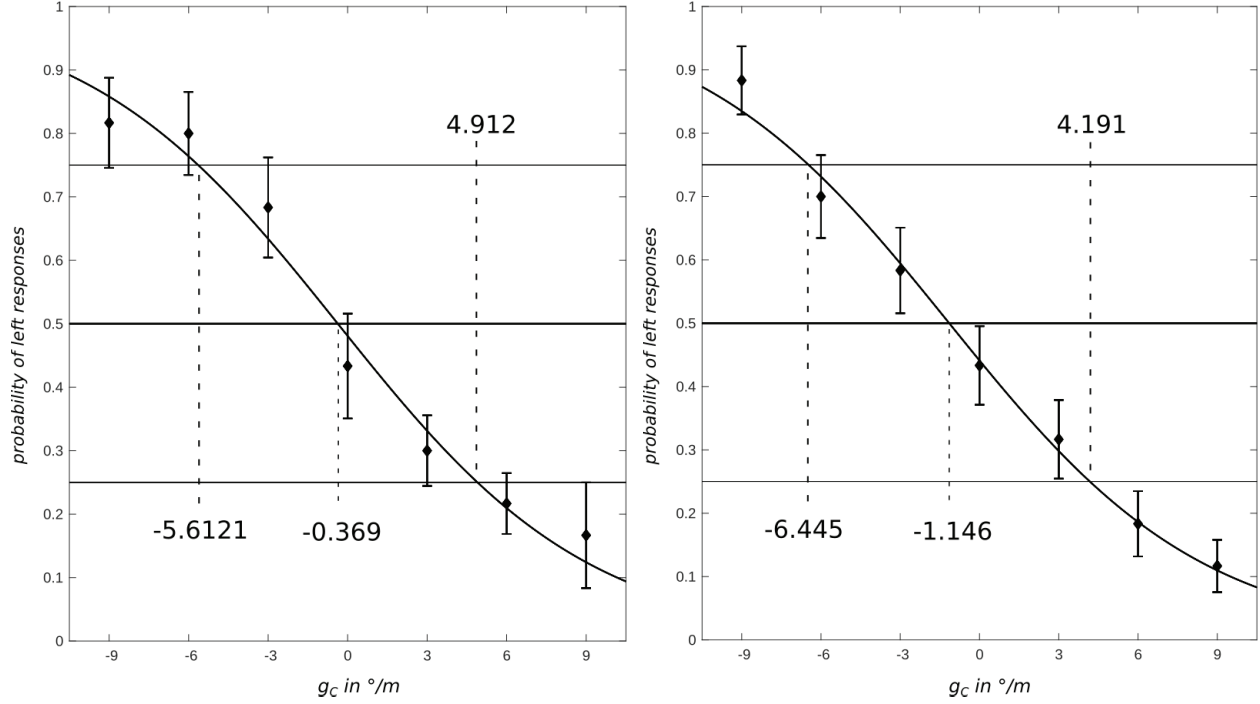


Figure 4: Psychometric curve that shows noticability of curvature gains for 360 video (left) and 3D modeled environment (right). DTs and PSEs are enlarged.

### 3.4 Results

#### 3.4.1 Detection Thresholds

Figure 4 shows the pooled results of the 2AFC task over all participants. The x-axis shows the applied gain in degrees per meter. The y-axis shows the probability of correctly detecting the manipulated path. For each gain, the mean and standard error bars are displayed. Each curve was fitted with a sigmoidal psychometric function which determines the DT. For 360 video, the lower DT is at  $-5.6121^\circ/m$ , the upper DT is at  $4.9116^\circ/m$  and the PSE is at  $-0.3690^\circ/m$ . For 3D model, the lower DT is at  $-6.4446^\circ/m$ , the upper DT is at  $4.1910^\circ/m$  and the PSE is at  $-1.1463^\circ/m$ .

Performing a significance test in order to compare the thresholds between 360 video and 3d model is not possible because it would require individual plots and DTs including standard deviation for each participant. However, calculating means of all individual plots might lead to different DTs than displayed in a plot with pooled results such as in Figure 4. Therefore, we decided to not conduct significance tests.

#### 3.4.2 VR Sickness

We measured a mean SSQ-score of 8.88 ( $SD = 11.08$ ) before the experiment, and a mean SSQ-score of 30.15 ( $SD = 34.34$ ) after the experiment.

### 4 DISCUSSION

The findings indicate that the detection thresholds of curvature gains in the 360 video environment fall within a similar range as the thresholds estimated in the 3D modeled environment. This outcome aligns with prior research on translation and rotation gains in 360 video environments [29], suggesting a certain degree of consistency in user perception thresholds across different types of VR manipulations.

Moreover, the estimated DTs are in a similar range as reported in comparable experiments such as  $4.9^\circ/m$  found by Grechkin et

al. [10] and  $5.2^\circ/m$  found by Rietzler et al. [20].

A noteworthy distinction emerged when comparing the two environments. As said, we did not conduct significance tests, so the following observations are not backed by statistical evidence and must be treated with caution. In the 360 video setting, the DTs and the PSE showed a more symmetrical distribution. In contrast, the 3D modeled environment presented asymmetrically shifted thresholds and a larger offset of the PSE, leaning towards the left side. Again, this confirms previous findings of the work by Zhang et al. [29]. Moreover, DTs of 3D modeled environments traditionally show an asymmetric characteristic of the psychometric function [25] that we could confirm an additional time.

This observation suggests that user perception and sensitivity to curvature gains tend to differ between these two environments, with users in the 3D modeled environment demonstrating a heightened sensitivity to curvature manipulations on one side of the spectrum.

Zhang et al. already discussed possible explanations for this behaviour. These include the characteristics of 360 video that were already mentioned in section 1: More realistic and detailed environments as well as lack of stereoscopic disparity and motion parallax. The better visual quality of 360 videos might lead to a higher weight of vision in multisensory integration and therefore less weight of vestibular and proprioceptive stimuli. This means lower sensitivity to detect manipulated visual stimuli. Our results support this theory. Consequently, this would also mean that stereoscopy and motion parallax are less important factors for the detection of redirection manipulations.

Still, it needs to be clarified why DTs in 3D modeled environments are shifted to the left. An explanation could be based on the majority of right-handed participants in these experiments who are more sensitive to redirections to their right side. However, this needs to be evaluated and confirmed in future work since we did not collect this data systematically within our experiment.

One limitation of our experiment lies in its transferability to

actual telepresence systems where a video stream from a remotely controlled maintenance robot, providing a dynamic and immersive view of the environment, is used.

Another limiting factor of our results is the gender imbalance among our participants that can make a difference [16, 28].

The increase of SSQ-score during VR exposure indicates an effect of the gains or the long exposure duration. It is known from previous studies that RDW can increase motion sickness. However, we do not know which gains exactly cause this because we did not evaluate SSQ-score per gain.

## 5 CONCLUSION

Our investigation into curvature gain detection thresholds within a VR telepresence system for remote robot control in hazardous environments yielded valuable insights. Notably, we found that curvature gain thresholds in 360 video and 3D modeled environments are in a similar range, aligning with previous findings for translation and rotation gains. However, a noteworthy distinction emerged: the 360 video environment exhibited symmetrical DT and PSE values, while the 3D modeled environment displayed an asymmetrical bias towards heightened sensitivity to curvature manipulations. Further research is necessary to more clearly find out about (dis-)advantages of the use of either 3d models or 360 videos.

In conclusion, our research advances our understanding of user perception in efficient redirected walking systems. These insights contribute to the ongoing development of VR telepresence and 360 video environments.

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